Data Mining Methodologies and Systems

- Structuring Unstructured Data for Knowledge Mining: A Data-Driven Approach
- Data Augmentation
- From Correlation to Causality
- Network as a Context
- Auto-ML: Methods and Systems

Network as a Context

- Network of X
  - Each node represents an entity, a data set or a data mining model
- Examples: Network of time series
  - The key idea for mining network of time series data
    - Model each time series by either traditional signal processing approaches or deep neural networks
    - Leverage the contextual network to regularize different time series models

Network as a Context

- Network of X
  - Each node represents an entity, a data set or a data mining model
- Examples: Network of networks
  - Each X (i.e., node) itself of the contextual network represents another domain-specific network
  - The main advantages of the network of networks model
    - Explicitly encodes the hierarchical structure
    - The contextual network provides additional regularization

Network as a Context

- Network of X
  - Each node represents an entity, a data set or a data mining model
- Examples: Network of data mining models
  - Each X itself is a data mining model
  - Different performance prediction models ‘borrow’ data from each other
  - Mutually boost each other’s prediction performance
- Future directions
  - The contextual network construction
  - Can be applied to more applications, e.g., knowledge graphs, heterogeneous networks
  - Integrate other data mining problems, e.g., adversarial learning, fairness, explainable learning.